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Mechanical Design of Roll Control Canards for a Sounding Rocket



Waterloo Rocketry

Waterloo ON, Canada

Benjamin Pickens

21058534

Mechatronics Engineering

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PD Evaluators
Centre for Work-Integrated Learning
University of Waterloo
Waterloo, Ontario
N2L 3G1

Dear PD evaluators,

I am writing to submit the report “Mechanical Design of Roll Control Canards for a Sounding Rocket.” This report will become or replace work term report 200. My previous academic term is 2A.

This report is written for a student design team: Waterloo Rocketry, of which I am a core member and subsystem co-lead of the Controls subsystem. Waterloo Rocketry launches a student-built rocket each year at competition and strives to train and provide experience for engineering students in the aerospace field.

The purpose of this report is to detail the design of the mechanical system for this year’s controls subsystem project: roll control canards. I was and am the lead on this design project along with now being a subsystem co-lead. Hopefully this report, along with the two other reports being written about this project: aerodynamic design by Luca and controller and simulation design by Finn and Joe, will provide valuable resources to future controls projects. It should also serve to assemble all the relevant information, particularly about decision making, of the mechanical aspect of this year’s project to help present the safety of our new and complex system to the competition judges.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I have checked this report for spelling and grammar errors to the best of my abilities. The content in this report is primarily my work but also includes the contributions of all Waterloo Rocketry team members, particularly those working on the controls subsystem. I would like to specifically thank Christine Li, my controls co-lead, Luca Scavone, flight dynamics lead and enabler of all controls work, Finn Breuer, simulation and controller project lead as well as extremely knowledgeable, Richard Ding, previous controls lead, Joe Dolina, other past controls lead and often saving us, and Tina, Jason, Tristan, Raag, all first years, for their incredible commitment to this project I have had the pleasure of working on.

Sincerely,
Ben Pickens
21058534

Executive Summary

This report presents the design, development, and testing of the mechanical mechanism for a new roll control system using aerodynamic canards created by Waterloo Rocketry's controls subsystem. The canards are the team's second major active controls project following airbrakes and represent both significant new development and continued learning from that project.

The main goals of the canards are to control the roll of the team's rockets, thereby providing stability and other benefits, as well as to further the development of active controls systems. The mechanical mechanism must mount two triangular aerodynamic control surfaces as well as control them while ensuring they remain mechanically linked. This is accomplished using three shafts connected by bevel gears with two of the shafts connecting to the control surfaces and the third to a servo. All of these components, along with the rest of the controls system including the battery for the servo and controls avionics, are then bolted to a baseplate which is then bolted to the rest of the rocket. This provides stand alone testing along with easy integration into both the TestFlight and Aurora rockets the canards are designed to fly on.

The canards were fully designed and manufactured in eight months and launched on the TestFlight solid rocket in May but were unfortunately accidentally disabled during that launch. They have since been slightly updated to integrate onto the Aurora rocket, set to launch in August. They have passed all validation steps to launch, and their existence represents a significant success for the controls subsystem. When they hopefully launch on Aurora they will allow testing of testing of the most capable simulation, rocket state estimation, and control software created by Waterloo Rocketry to date.

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1 Introduction

This report will detail the design of the roll control canards system designed for the 2024-2025 Waterloo Rocketry sounding rockets: TestFlight and Aurora. The canards are a brand-new system designed to solve a problem new to the team but using some principles and techniques learned from last year's controls project: airbrakes. This report will specifically cover the mechanism that actuates the canard aerodynamic control surfaces and there are other sources of documentation for the aerodynamic design, simulation/estimation/controller design, and firmware (implementation of control software) design.

2 Background Information and Problem Description

2.1 Waterloo Rocketry

Waterloo Rocketry is a student team at the University of Waterloo that competes annually in the Launch Canada advanced category since August 2024 and competed at the International Rocket Engineering Competition (IREC) for many years before that (Waterloo Rocketry, 2025). This year the team is attempting to launch Aurora, a bipropellant liquid rocket fueled by nitrous oxide and ethanol. It will fly to a target height of 50,000 ft and reach a top speed near Mach 2.5 (2.5 times the speed of sound). The rocket is an 8in cylinder for most of its length with fins at the back and a conical nosecone at the front. At the time of writing of this report the rocket is expected to be 17ft long (Waterloo Rocketry, 2024). Below is an image of last year's rocket Borealis.



Figure 2-1 - Borealis on the launch rail

2.2 Last Year's Project: Airbrakes

Last year Waterloo rocketry launched its first active control project, airbrakes, and founded the Controls subsystem. The airbrakes project used small panels extended horizontally from the rocket to create controlled amounts of drag and therefore slow the rocket and control the final height (apogee) it reaches. Significant technical development was completed for this project that was reused or influenced the design of the canards. The main control printed circuit board (PCB), was remade with only slight modifications, the servo actuator and the mechanism by which it was mounted were reused, and importantly for this report the overall architecture of the controls system and its integration were reused. This consisted of mounting all the mechanism components to a circular plate then mounting that plate to the rest of the rocket. Below is a screenshot of the airbrakes assembly in computer aided design (CAD).

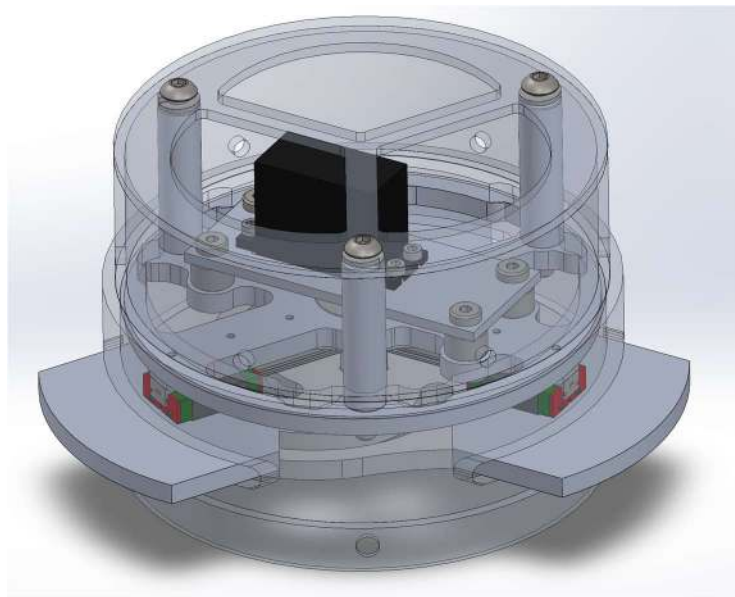


Figure 2-2 - Screenshot of airbrakes assembly, figure is of a file created by the author, Ben Pickens, stored locally

Unfortunately, the airbrakes did not have the opportunity to prove their functionality during flight as the rocket undershot the programed apogee target and therefore the airbrakes would not

have activated as they can only slow the rocket to achieve a lower apogee. It was discovered after the flight logs were analyzed that they were also deactivated and would not have worked even if the flight had gone high enough. Regardless, most of the project milestones were completed, and the system worked in testing, so it was considered to be effectively completed enough for the team to move to a new project.

2.3 Why do Canards Exist

High altitude sounding rockets experience a variety of vehicle dynamics problems including a specific one called coning (Howard L. Galloway, 1966). Coning happens as the damping force from the fins becomes too weak to keep the rocket pointed straight up against the influence of roll pitch coupling (Howard L. Galloway, 1966). Roll pitch coupling is where the roll rate matches the natural pitch frequency and any small pitch moments from imperfections in the fins, misalignment of the center of gravity, or any other source become amplified until the rocket is flying at a high angle of attack, where the rocket long axis is not pointed straight along its path of motion (Howard L. Galloway, 1966). At a minimum this causes significantly reduced apogee due to the increased drag, but if it is too severe the rocket can break up under the increased loads (Howard L. Galloway, 1966). Importantly one of the best ways to control this is to ensure the rocket does not roll at or near the natural pitching frequency (Howard L. Galloway, 1966).

Aurora is unlikely to experience a significant impact from this effect as its flight will mostly stay within thicker atmosphere where the damping provided by the fins will help prevent it (Howard L. Galloway, 1966). However, as the Waterloo Rocketry rockets continue to grow larger and go higher, this effect is expected to become a significant risk. Therefore, in preparation for those

flights, and to develop technical expertise in active flight control systems the controls subsystem decided to select effective roll control as the design problem to solve this design cycle.

2.4 Active Attitude Controls on Amateur Rockets

Unfortunately, because of the difficulty in creating active attitude control systems, systems that control the direction and angular motion of a vehicle, for high powered rockets and the export-controlled nature of some of the information involved, there are limited similar projects accessible online to reference. This significantly increased the difficulty of designing the canards compared to the airbrakes which, because of the specific niche of student rocket competitions, could reference many other reports about projects with the same goal. Two systems published online that were referenced for the canards project were “Development of a Canard Based Vertical Orientation System for High Power Rockets” by Jim Jarvis and “Building a Guided Rocket to Hit Mach 3” by Joe Barnard on the BPS.space YouTube channel. Both projects were creating full 3 axis attitude control systems, controlling the direction the rocket is pointed and its roll, while the canards project is only designed to control roll however much of the modeling and simulation information was useful reference. Unfortunately, neither project created a mechanical system which could meet the integration constraints of the canards project and therefore neither provided mechanical design references.

2.5 TestFlight

At the beginning of this design cycle, in September, the team decided to explore the possibility of launching a test rocket with the intention of providing specific subsystems with the opportunity to launch more than once per year and specifically this year to test riskier aspects of the controls, recovery, and payload subsystems on a less expensive and lower stakes launch. The main design requirement for this “TestFlight” rocket was simplicity and minimal required work.

Therefore, the rocket was constructed with commercial body tubes, and a commercial solid motor was purchased for it. The vehicle architecture above the motor was intentionally similar to the planned Aurora vehicle architecture and therefore the canards system was able to be designed to mostly meet the requirements of both launches. TestFlight launched May 10th with a fully nominal flight except for the canards active control software which was disabled by a timer lockout overlooked in the operations planning.



Figure 2-3 - The TestFlight rocket on its launch rail before liftoff with the canards visible, photo taken by author

The TestFlight launch schedule significantly compressed all aspects of the timeline for the canards project from a full year to eight months. This significantly increased the difficulty of the project and forced many decisions to be made based on what was fastest not what was best.

3 Mechanical Design Process and Details

This section of the report details various aspects of the design process and final design including both the results and discussion of those results for each aspect.

3.1 System Location - Injector section vs. Above Tanks

At the beginning of the canards project two options for the location of the canards within the rocket were proposed. They could either go in the vent section, immediately above the tank and in front of both the center of pressure (CP) and center of gravity (CG) of the rocket, or they could go in the injector section aft of both the CP and CG (Scavone, 2025). Both sections can be seen below in Figure 3-1 which shows the full layout of Borealis and is very similar to the layout of Aurora. Each of these options had tradeoffs and created different risks and requirements. They also created different amounts of work required to be completed by different groups within the team and this massively complicated the decision-making process.

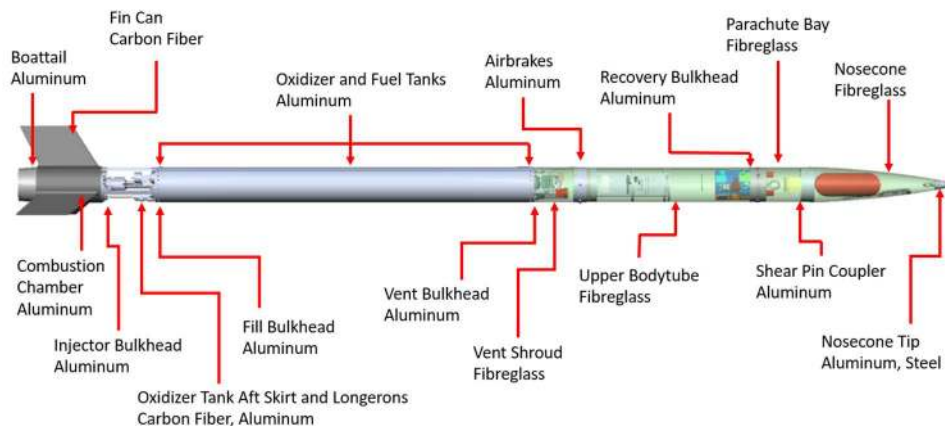


Figure 3-1 - Aerostructures layout of Borealis (Waterloo Rocketry, 2024)

Placing the canards in the injector section was originally determined to remove both significant risk and scope of work from the aerodynamic analysis required for this project. This option was therefore advantageous to the flight dynamics subsystem and people working on those aspects of

rocket design. However, selecting this option would require extremely difficult integration of two complex systems into one small space. Both the canards system and all of the plumbing, electrical, and actuation components normally required in the injector section would need to fit around each other and pack into an eight-inch circle.

Placing the canards in the vent section was determined to create effectively the opposite requirements. The mechanical integration scope would be significantly reduced as the canards could exist in a free eight inches of space and connect to the relatively empty top surface of the vent bulkhead. The aerodynamic analysis would then get significantly more difficult as with the canards in front of the CP and CG the aerodynamic stability of the whole rocket would be negatively impacted by the existence and especially movement of the canards.

After lengthy discussion involving the major stake holders it was initially decided to place the canards in the injector section to minimize the aerodynamic analysis required at the expense of increased mechanical integration difficulty. This was thought to be the preferred option as the team generally has more experience with mechanical design compared to most other aspects of building and launching rockets. During and after this decision-making period several concepts were proposed to integrate mechanically linked canards into the injector section. One of the more developed concepts can be seen in Figure 3-2 where a single large gear sits around the oxidizer feedline running through the middle of the section. Smaller bevel gears then connect to it for each of the canard shafts and the servo shaft. This solution solves one of the main problems with integration into this section: fitting the canards mechanical connection around the oxidizer feedline.

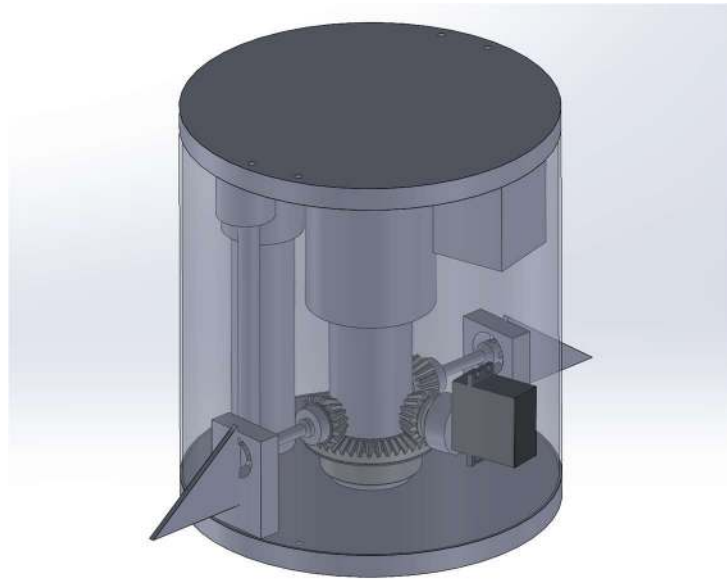


Figure 3-2 - Central ring gear concept for injector section canards, figure is of a file created by the author, Ben Pickens, stored locally

However, four months into the project, there had been minimal progress on either of the two complicated systems, both of which were required to fit into the injector section. Therefore, after further research into the aerodynamic analysis, see (Scavone, 2025), it was decided to switch the location of the canards to the vent section. This also allowed the canard system to integrate in a nearly identical manner between the TestFlight rocket and Aurora removing the requirement to create a mock injector section for TestFlight and remake more of the canards for Aurora. This was a critical decision and based on the tight timelines imposed by the TestFlight launch date was necessary to complete the system on time. The scope of work required to develop, integrate, and manufacture two systems with such tight integration requirements in a tight timeline was thoroughly underestimated and this project narrowly avoided becoming an expensive, sad example of that.

3.2 Final Design for TestFlight – Details and Select Analysis

The final design for the mechanism of the canards incorporates 3 main shafts geared together using bevel gears as shown in Figure 3-3. This design was specifically created to meet the requirement of mechanically linking the canards together so that they can never actuate separately. These three sub-assemblies along with the other components such as the battery mount, camera mounts, and canards avionics are then bolted to a single main baseplate. This allows for full integrated testing of the system without requiring other rocket components as well as significantly easier integration into any launch vehicle this assembly is launching on.

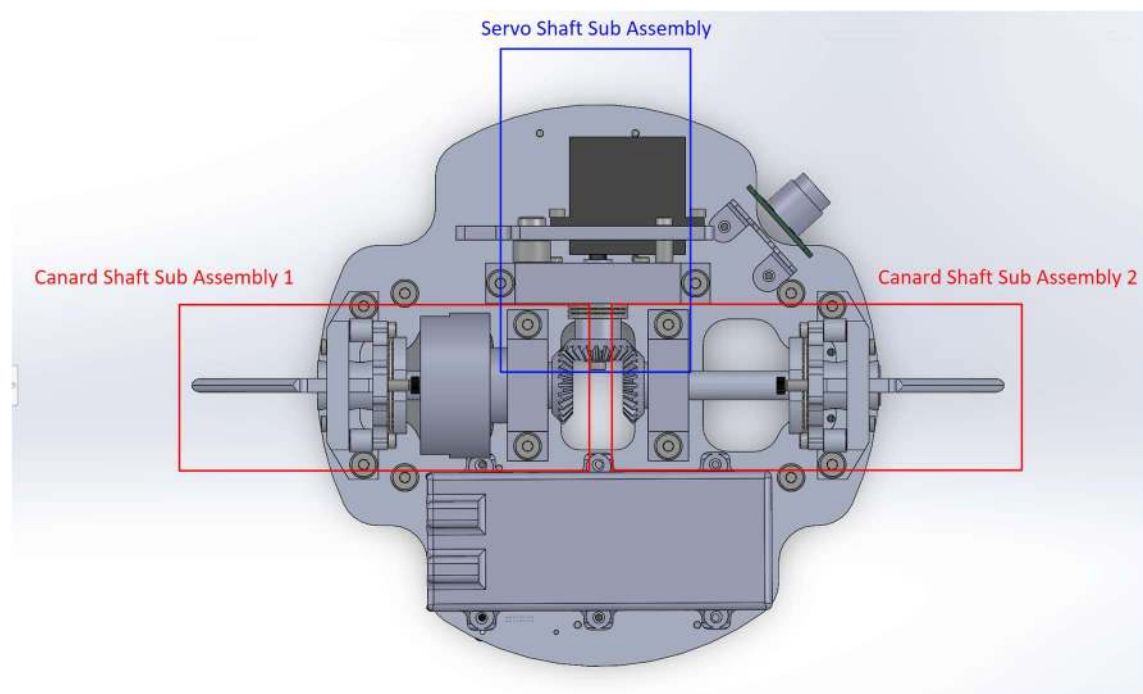


Figure 3-3 - Canard mechanism top view, figure is of a file created by the author, Ben Pickens, stored on team drive

The canard shaft sub assembly has several main functions. It provides secure mounting for the canard control surfaces, maintains the exact position of the gear to minimize backlash, and connects the two control surfaces rigidly. A through hole potentiometer is also mounted on one of

the canard shafts with a set screw. To retain the shaft a mounting block with a bearing was placed at each end. The outer end uses an ultra-thin bearing to fit around the control surface mounting geometry while the inner end uses a standard bearing to save cost. Each mounting block is positioned onto the baseplate using locator pins, although because of the difficulty in retaining them some were removed after assembly. The full sub assembly can be seen in Figure 3-4.

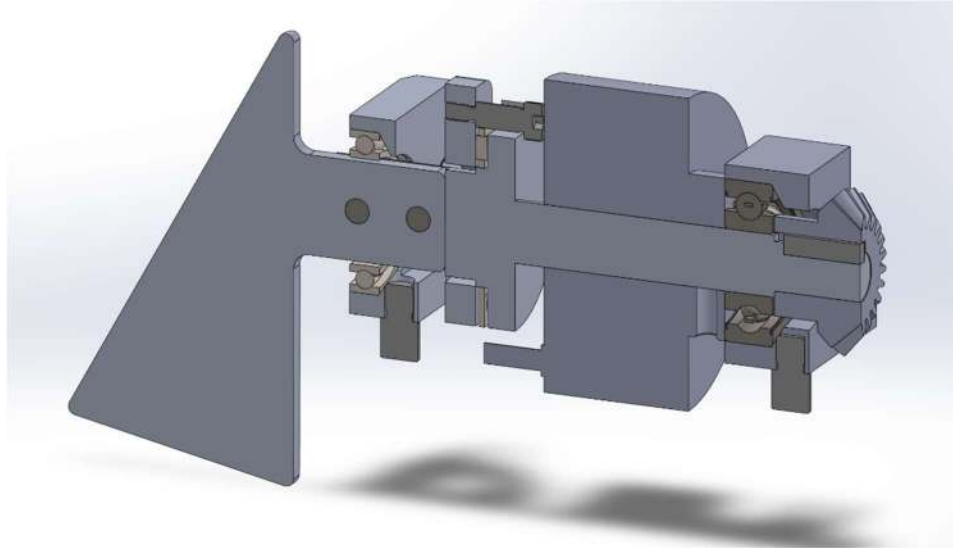


Figure 3-4 - Canard shaft sub-assembly section view, figure is of a file created by the author, Ben Pickens, stored on team drive

To mount the canard aerodynamic surfaces, two shoulder bolts pass through one side of the shaft, the canard shaft, then the other side of the canard shaft and into a tight space nut. This both locates the canards repeatably using the shoulder and fixes them using the shoulder and clamping forces. This, along with the canard profile, including the tab to connect to this mounting solution can be seen in Figure 3-5.

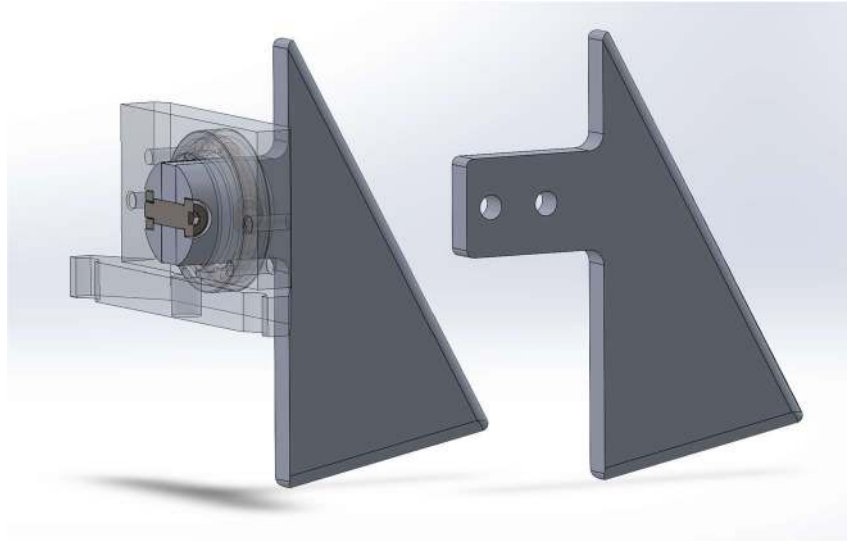


Figure 3-5 - Canard aerodynamic surface mounting and profile, figure is of a file created by the author, Ben Pickens, stored on team drive

To control the backlash of the gears and therefore minimize and negative impacts the backlash could have on the control loop stability a shim system was used. Figure 3-6 shows the force path through which the gear is located and how changing the number of shims therefore changes the axial position of the gear. Various numbers of 0.002-inch shims can be used with the nominal number being five.

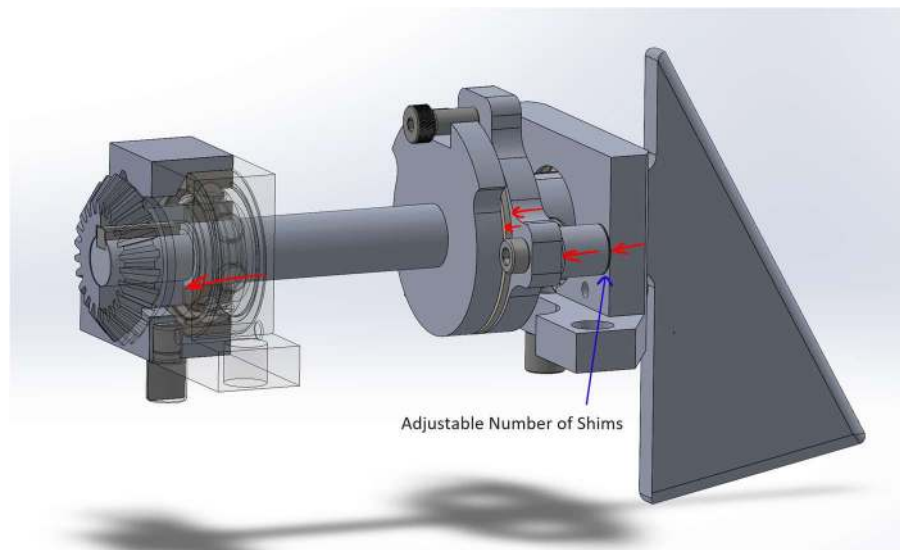


Figure 3-6 - Shim adjustment force path, figure is of a file created by the author, Ben Pickens, stored on team drive

The servo shaft sub-assembly, depicted in Figure 3-7, is similarly structured with a shaft running through a bearing and mounting block to a gear at the center. The servo shaft has only one mounting block because it is significantly shorter and because the bushing in the servo also helps maintain its position. The shaft was turned down from a hex servo shaft to allow it to both connect to the servo and fit through the specific diameter parts on it.

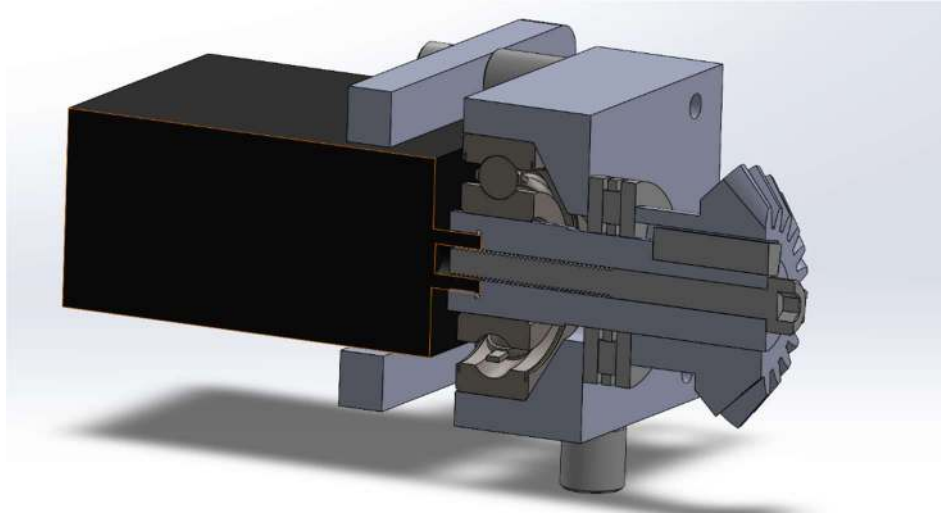


Figure 3-7 - Servo shaft sub-assembly section view, figure is of a file created by the author, Ben Pickens, stored on team drive

3.3 Select Examples of Analysis Performed to Create and Validate TestFlight Assembly

Here are two examples of analysis performed during the design process to ensure the canards could be both manufactured effectively and meet the functional requirements set for them. These are included to lay out a thought process which could provide an example for another design as well as to explain decisions made for this design. These were also two specific aspects of the design considered critical to analyze, and as such, they were among the few to receive a full analysis within the TestFlight timelines. Most other systems were simply overbuilt for TestFlight then analyzed more completely in preparation for the Aurora launch.

3.3.1 Keyed Shaft Calculations

One aspect of the design considered as a potential mechanical failure point was the canard shafts at their connection to the gear. This connection was identified as a weak point due to the small effective areas needing to transmit the full torque of the servo. Two particular failure modes were analyzed: the shear stress due to torque on the canard shaft and bearing stress from the key being pushed into the keyway cut into the shaft. The areas relevant to each of these failures can be seen below in Figure 3-8.

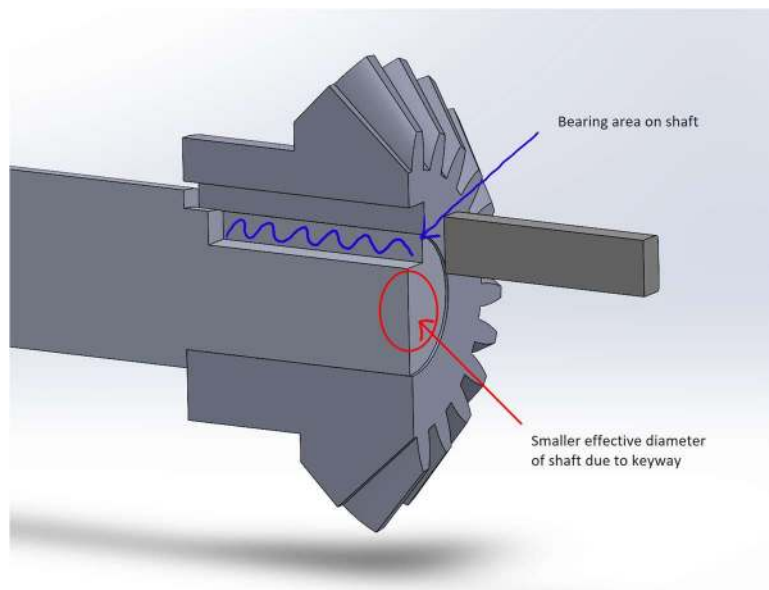


Figure 3-8 - Effective areas for failure modes of keyed shaft figure is of a file created by the author, Ben Pickens, stored on team drive

To analyze these stresses the following tables were created in a google sheet which can perform the necessary calculations given different input values. Table 3-1 contains all of the input values and the source of those values. Any reference to McMaster-Carr is the dimension of certain components, while a table of keyway dimensions was used from Hallite for the depth of the shaft keyway. The math used is concepts of the mechanics of deformable solids and can be found in

(Budynas, Nisbett, & Shigley, 2021). The following link provides access to the spreadsheet:

https://docs.google.com/spreadsheets/d/1lQahw_vF0g1lwwkcj8BsPRs7qP5YuL8ZyOu0agddaS0/edit?usp=sharing.

Table 3-1 - Input values for stress on shaft due to torque

Input Values		original unit		calculation unit	Source
Square Key Dimension	0.125	in	3.175	mm	(McMaster-Carr, 2025)
Depth of Shaft Keyway (B)	0.075	in	1.905	mm	(Hallite, 2017)
Outer Diameter - Original	0.25	in	6.35	mm	(McMaster-Carr, 2025)
Diameter at Bottom of Keyway - Original	0.1	in	2.54	mm	Geometry of Part
Outer Diameter - New	0.3125	in	7.9375	mm	Design Decision
Diameter at Bottom of Keyway - New	0.1625	in	4.1275	mm	Geometry of Part
Applied Torque	50	Kg cm	490	Nmm	Servo Rating
Shear Strength Aluminum	207	MPa	207	MPa	(ASM Aerospace Specification Metals Inc., 2024)

Table 3-2 Results of stress on shaft due to torque

Results	J (mm ⁴)	Max Shear Stress	FOS
J Outer Diameter - Original	159.62	9.75	21.22
J Inner Diameter - Original	4.09	152.39	1.36
J Outer Diameter - New	389.70	4.99	41.45
J Inner Diameter - New	28.49	35.51	5.83

Based on these calculations it was determined that the shaft with the original diameter designed to fit into the unmodified gear could fail under the torque applied by the servo. The analysis was

known to be conservative by not counting the full area of the shaft and instead only including the area within the depth of the keyway. However, given the critical nature of the canards system it was decided to simply drill out the center of the gear and increase the diameter of the shaft instead of attempting more complex analysis which could contain errors.

3.3.2 Thrust Bearings Selection and Implementation

To select the correct thrust bearings for retaining the canard shafts both the geometry constraints and loads were analyzed. This was initially a particularly difficult problem to solve while maintaining easy assembly but the final solution ended up using a needle roller thrust bearing sandwiched between a flange on the shaft and a separate plate stood off from the outer mounting block as can be seen below in Figure 3-9. The geometry constraints therefore required a thrust bearing with a large inner diameter which left several choices: needle-roller thrust bearings, cylindrical roller thrust bearings, tapered-roller thrust bearings, and thrust ball bearings (McMaster-Carr, 2025). To decide between them the cheapest one which could withstand the required thrust load was selected. The thrust load was determined by using the relationships between torque and thrust load for bevel gears. For a 90-degree bevel gear, a gear designed to interface with another gear at 90 degrees to it, the ratio is 1:1 (Budynas, Nisbett, & Shigley, 2021). Therefore with a torque of 50 kg cm or 43.4lb in, the maximum torque transmitted by the servo, and a gear with a 1 inch pitch diameter, 43.4lb of axial load will be created (Budynas, Nisbett, & Shigley, 2021). Therefore, the smallest and cheapest thrust bearing available which fits the constraints is the needle roller thrust bearing, as the thrust ball bearing can only withstand 45lb and this was decided to small of a safety factor (McMaster-Carr, 2025).

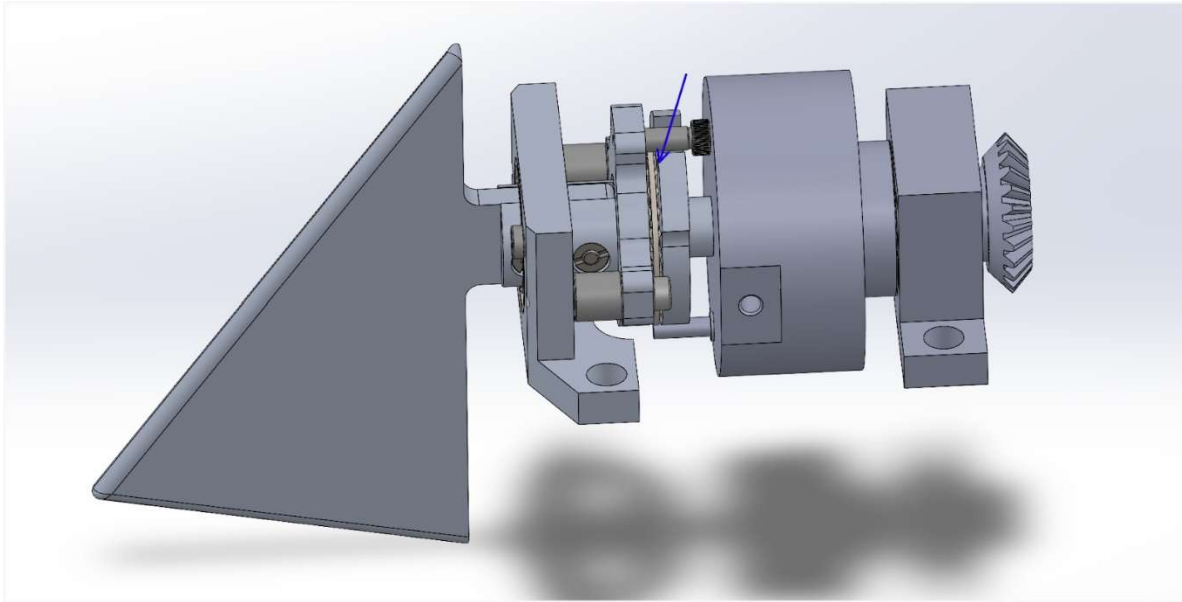


Figure 3-9 - Final thrust bearing selection and location in assembly, figure is of a file created by the author, Ben Pickens, stored on team drive

4 Modifications for Aurora Launch Requirements

Broadly the canard system remains unchanged between the TestFlight configuration and Aurora configuration. The changes were minimized for several reasons. Even though the control loop was accidentally disabled during the TestFlight launch and therefore the mechanism actuation is untested on flight all other aspects of the system were verified. This, combined with ground testing of the actuation, makes the canards almost flight proven and therefore easier to present to the competition and prove the safety of. Minimizing the changes also reduces the total cost and required work for the full design cycle, while allowing for more in-depth analysis and testing at a less rushed pace.

Some changes, however, have been required. The control surfaces are swapped for smaller ones to account for the increased air speed of Aurora and keep the control authority similar (Scavone, 2025). The final dimensions are shown in the second triangle of Figure 4-1 with units in inches. The inner diameter of the rocket body tube the canards fit into is slightly smaller than on TestFlight and therefore some components of the canards have been filed or otherwise slightly modified to fit within the new tube. Camera mounting has also been updated to fit around components of the vent section with which the canards now integrate.

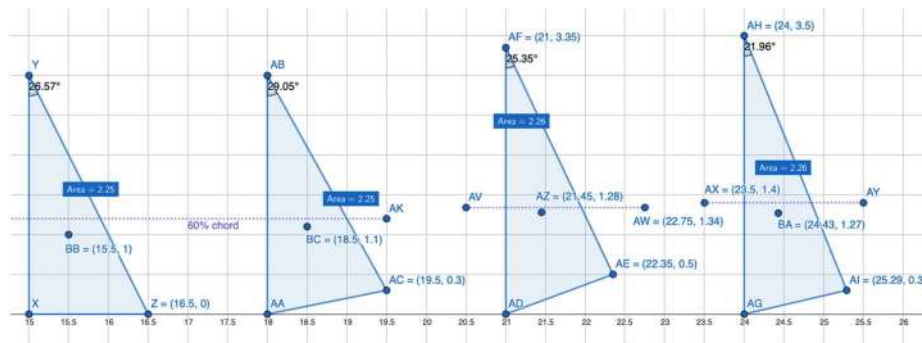


Figure 4-1 - Proposed canard designs (Scavone, 2025)

5 Conclusions and Recommendations

Overall, the canards mechanism succeeded at being manufactured, assembled, benchtop tested, and flown on TestFlight. These are all significant successes. It is now being modified to hopefully fly on Aurora and enable testing of the most capable simulation, rocket state estimation, and control software created by Waterloo Rocketry to date. Regardless of the success of the canards on Aurora, their existence now represents a massive accomplishment.

Not all aspects of the process went smoothly however, and several important lessons were learned. Some of the key high-level takeaways are:

- Designing both novel complex systems and novel complex integration solutions at the same time will always be significantly more difficult than predicted. Any situations that fit this description should be avoided whenever possible. It is much easier to extend the timeline and create system, test and understand it, then integrate it or something based on it into any complicated integration requirements.
- Communication and documentation become significantly more critical the more integrated a system must be.
- Iterative design focusing on critical functionality first, then less critical functionality like mounting is extremely effective and the pace this project was completed with would not have been possible without those strategies.

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